CS 532, Spring 2019: Quiz 3

NAME:

Problem 1. Consider two images of the same irregular five-sided polygon, which are related by an unknown homography. Assume there is no other image content and that you can reliably detect corners in both images. In order to estimate the homography transformation between the two images you make an exhaustive list of all putative correspondences between corners. Answer:

- How many putative correspondences do you have?

- What is the inlier rate?

- How many RANSAC iterations are needed to reliably estimate the homography (Approximate values or formulas are sufficient) ?

- Does the problem change if the shape is a regular (i.e. equilateral) pentagon? Discusss.